

# L12: Maximum Likelihood Estimation of HRG Probabilities

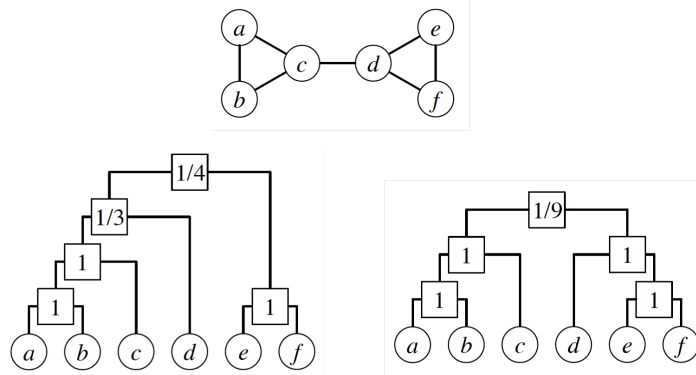
In the previous page, we assumed that the dendrogram  $D$  and the vector  $\{p_r\}$  of the HRG model are given. In that case, we can easily create many random graphs  $G$  that follow the corresponding HRG model.

What if we do not know these parameters but we are given a (single) network  $G$ , and our goal is to statistically estimate the parameters  $D$  and  $\{p_r\}$  so that the HRG model can produce graphs that are similar to  $G$ ?

Let us first show how to calculate the probabilities  $\{p_r\}$  from a given network  $G$ , assuming that the dendrogram  $D$  is given (we will remove this assumption at the next page).

We will do so using the well-known **Maximum Likelihood Estimation (MLE)** approach in statistics.

Suppose that  $r$  is an internal node in  $D$ , and consider all node pairs for which  $r$  is the LCA. Let  $E_r$  be the number of edges in  $G$  that connect those node pairs.



In the left dendrogram of the visualization, the internal node  $r$  that is labeled as “1/3” is the LCA of the following node pairs:  $(a,d)$ ,  $(b,d)$ ,  $(c,d)$  – and only one of those node pairs is connected (c with d). So  $E_r=1$ .

Further, let  $L_r$  and  $R_r$  be the number of leaves in the left and right subtrees rooted at  $r$ .

In the previous example,  $L_r=\{a,b,c\}$  and  $R_r=\{d\}$ .

Suppose that someone gives us a vector of probabilities  $\{p_r\}$ . Recall that the likelihood of a model is the probability that that model will produce the given data. So, the model likelihood is:

$$\mathcal{L}(D, \{p_r\}) = \prod_{r \in D} p_r^{E_r} (1 - p_r)^{L_r R_r - E_r}$$

with the convention that  $0^0 = 1$ .

So, if  $D$  is fixed, what is the vector  $\{p_r\}$  of the HRG model that will result in maximum likelihood?

If we differentiate the previous expression with respect to  $p_r$  and set the derivative to zero, we can easily calculate that the optimal value of each  $p_r$  is:

$$\hat{p}_r = \frac{E_r}{L_r R_r}$$

which is simply the fraction of potential edges between the two subtrees of  $r$  that actually exist in  $G$ .

For these optimal values of  $p_r$ , we can easily show that the model likelihood is:

$$\mathcal{L}(D) = \prod_{r \in D} [\hat{p}_r^{E_r} (1 - \hat{p}_r)^{L_r R_r - E_r}]$$

It is common to work instead with the logarithm of the likelihood:

$$\log \mathcal{L}(D) = - \sum_{r \in D} L_r R_r h(\hat{p}_r)$$

where  $h(p) = -p \log p - (1 - p) \log(1 - p)$ .

As an exercise, use this last formula to show that the likelihood of the left dendrogram shown in this visualization is approximately 0.00165. Then, show that the likelihood of the right dendrogram is higher (approximately 0.0433).

### Food For Thought

Write down the derivations of this page in more detail yourself.